

DC ARMOUR REFERENCE

Screen card. Distilled from 11 tutorials — [forge/ledger/](#)

WHAT THIS IS

Eleven armour/asset tutorials transcribed in full and boiled down to the moves that transfer. Two tracks, kept separate on purpose: ZBrush leads, Blender runs alongside.

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HOW TO READ IT

Nothing here is recalled. Every setting traces to a transcript in [forge/ledger/raw/](#). Where a source states a number it is reproduced as stated; where none was stated, none was invented.

THE THREE LAWS ALL SOURCES AGREE ON

1 · BUILD IT THE WAY IT WAS BUILT

Separate pieces, real seams, real overlaps. Never sculpt a multi-part object from one blob. Decide splits from how the thing is WORN.

2 · BIGGEST SHAPES FIRST

Block the body before any piece; nail the primary form before any panel line. If the scale is wrong, fix the big shapes — never compensate downstream.

3 · PERFECT READS FAKE

Break the surface on purpose so it reads hand-hammered, not machine-made. Deliberate dents, deliberate asymmetry, deliberate wear.

NO TIMESTAMPS — WHY

YouTube's caption API refuses this network (IpBlocked), and the route that worked returns untimed text. So this is organised by technique, not timecode. To find a moment on screen, `grep forge/ledger/raw/` for the phrase.

ZBRUSH — ARMOUR WORKFLOW

Gatz 3D (the spine) · TriGon (the full build)

THE FIVE STAGES, FOR A WHOLE HARNESS

- 1 BLOCK OUT the character. Proportions and spacing correct before anything else.
- 2 3D SKETCH every piece straight onto the model. Loose and fast — not the detail phase. Duplicate and HIDE the base mesh first.
- 3 SPLIT every piece off: mask → Polygroups > Group Masked → Split Masked Points. ~50 pieces is normal.
- 4 THICKEN each piece (right column).
- 5 REFINE: no gaps, silhouette right, proportions right.

Why this order: it stops you finishing one pauldron while the rest of the harness is nothing. Work everything a little.

FIRST SHAPE — the extract route

mask	lasso for the gesture, pen for precision. Don't fuss.
Extract	smoothness 0, thickness 0, Tab Border OFF → Accept
then	Polish By Features → ZRemesher LOW → Polish again
then	Inflate slightly, Auto Groups, Delete Hidden

THICKEN — the transpose method

Ctrl+Shift+clk	isolate the piece
Close Holes	Geometry > Modify Topology — fills the back
Transpose on drag centre	Ctrl+click the FRONT polygroup = masks the rest of the three rings → extrudes it
Inflate	Deformation > Inflate recovers over-thinning

Faster for simple/repeatable pieces: mask → Subtool > Extract at 0.001. Many limb pieces are the same shape — build one, duplicate, move.

DESIGN EXPLORATION — Dynamesh is the thinking tool

thickness	make it THICK. Thin creates holes. Offset -100 if needed.
Dynamesh	LOW res — low enough to shove, not so low the form dies
EQ Curve	Brush > Curve. OFF = round pulls. ON = POINTED pulls.
Render>Flat	kills shading — judge pure silhouette
Shift+S	screenshot variants and compare side by side

The armour's job is to BREAK the silhouette away from the body. A near-straight line mirroring the torso is the failure state. Stagger the high points — equal heights and equal strap gaps read boring.

ZBRUSH — HOTKEYS & MENU PATHS

Only what the sources state. Nothing guessed.

SELECTION / VISIBILITY

Ctrl+Shift+click	isolate piece, hide rest
Ctrl+W	group visible into one polygroup
Auto Groups	polygroup per connected piece
Delete Hidden	commit a visibility cut
Split Hidden	hidden part → own subtool
Visibility>Grow	expand / Shrink contracts

MASKING

Ctrl+drag	mask · Ctrl+Alt+drag unmask
Ctrl+click bg	invert mask
Ctrl+Alt+click	SHARPEN the mask — used constantly
Mask By Features	mask the polygroup/edge features
Masking>Cavity	then invert + sharpen + Inflate = fibres
backface mask	Brush > Auto Masking > front faces only

BRUSH MODIFIERS

Alt (held)	invert; 2nd pass on DamStandard sharpens it
Ctrl+drag IMM	place repeats at a LOCKED size
Brush>Depth	how deep an IMM sinks (negative = in)
AutoMask>Range	stops Move affecting the far side (~7)
Lazy Mouse	on for any long clean stroke

GEOMETRY / TOPOLOGY

Dynamic Subdiv	live thickness + smoothing preview
Crease PolyGroup	Geometry > — crease on group borders
Group By Normals	~70 — polygroup by angle
ZRemesher	+ KEEP GROUPS = follows your polygroups
Polish By Feat.	THE workhorse cleanup. Mask corners first.
Dynamesh + Groups	keeps polygroups from merging
Project History	Subtool > Project — re-project earlier state
Extract 0.001	fast piece from a mask
Merge Down	keeps subdivs if levels match

STROKE / CURVES

Frame Mesh	Stroke > — by Border, or by creased edges
Curve Mode	Stroke > — set Dots if it won't snap
Curve Step	spacing along a curve. 0.9 = tight.
Create InsertMesh	Brush > — perspective OFF, view flat on

KNOWN FLAKINESS — from the sources

Live Boolean refuses → duplicate the subtool. Slice Curve ignores symmetry → Mirror And Weld after, every time. Mirror And Weld misbehaves with Dynamic Subdiv on. Stuck op → Ctrl+click the timeline. SAVE before any boolean.

ZBRUSH — BRUSH ROSTER

What each one is actually for · Hristian Shyne, TriGon

METAL SURFACE

Flatten	the main metal brush. High strength + metal 01 alpha. Facets the surface. It STOPS — it won't punch through. Hits backfaces: mask them.
hPolish	flattens, cleans, and breaks up noise that came out too uniform.
Clay Buildup	alpha 18 + spray, RGB off, ZAdd 2–3 = hammered metal. alpha 40/47 spray = fine dots.
Slash	low intensity — edge damage and jagged separation.
DamStandard	scratches and cracks. 2nd pass holding Alt sharpens. Transparent ON draws THROUGH a mesh.
Trim Dynamic	edge breakup alternative.
Snake Hook	pull individual stitches loose.

THE LAYER DISCIPLINE — non-negotiable

new layer	every damage pass gets one = one intensity dial
morph target	STORE it = selective undo for that pass
then	sculpt HARDER than you want, dial down after

You cannot judge coverage at low intensity. Work visible, then tone down. Bake layers when a pass is done — live layers bloat the ZTL.

THE NOISE STACK — multiple scales, each on a layer

- 1 Surface Noise at a LOW subdiv, strength very low → hammered dents.
- 2 Surface Noise at the HIGHEST subdiv, lower still. 'I barely want to see it.'
- 3 NoisePlug set to UV not 3D — stops it tiling perfectly, feels natural.
- 4 One very large noise at the LOWEST subdiv (~0.2) to warp the plate.

Mask it so it isn't uniform: Mask By Smoothness / Peaks and Valleys, paint by hand, then morph detail back out. Save presets — 'metal noise 1, 2'.

BAKE-FORWARD — design for the bake

Break up the edge-to-plate junction so curvature/AO catches it. Give plate / ornament / edge / bolts SEPARATE POLYGROUP COLOUR IDs so Substance can split materials. Always ask how you'll retopo it before you detail it.

BLENDER — THE SHRINKWRAP PANEL METHOD

Baril3D · the one worth automating ([forge/panel_stack.py](#))

THE IDEA

A BASE MESH carries the form. A WORKING PANEL carries the detail and is shrinkwrapped onto it. Now the panel's topology is FREE — cut loops, inset, punch holes — and curvature stays perfect, because curvature lives on the base.

STACK ORDER — it matters

Mirror	clipping ON
Subdivision	level 2 · Advanced > KEEP CORNERS
Shrinkwrap	target = base copy · group 'wrap'
Solidify	~3 mm
Bevel	small, 1 segment
Weighted Normal	Auto Smooth sits ABOVE this

THE BASE MESH COPY

Subdivision 5	(4 if slow)
Solidify	DISABLED — a solidified edge corrupts projection
hidden	viewport + render

APPLY SCALE before Solidify or thickness is wrong. Start from ONE vertex and extrude the silhouette — low counts stay editable.

VERTEX-GROUP OFFSETS — the powerful part

'wrap' all verts → Shrinkwrap 1 (normal projection)

'offset' a subset → Shrinkwrap 2:

Mode = OUTSIDE SURFACE · Offset = 3 mm · Vertex Group = offset

Result: a rolled sheet-metal lip standing proud of the surface — fully re-tunable, and still flat in edit mode. Stack a third for 6 mm, or set it INSIDE for recessed rivet beds. Remove verts from the group to soften falloff.

RIVETS

snapping	magnet ON, snap to FACE
tick BOTH	Project Individual Elements + Align Rotation to Target
Alt+D	LINKED duplicate — edit one, edit all
pivot	Individual Origins to scale them independently

HOLES — LoopTools route

Select face row → Select > Checker Deselect → Inset → right-click > LoopTools > Circle → pivot Individual Origins → scale down → delete faces → extrude inward for a rim. Tip: INSET TWICE up front and skip the extrude.

LATTICE — last resort, and last in the stack

Select parts, Shift-select lattice LAST, Ctrl+P > Lattice Deform. Afterwards the mesh 'pops off' in edit mode. Use only at the end to fix proportions.

BLENDER — HOTKEYS

From the four Blender sources only

TRANSFORM & SELECT

G twice	EDGE SLIDE along the surface
S X 0	flatten a selection onto an axis
S then Shift+Z	scale on X and Y only
Ctrl+A	APPLY TRANSFORMS — before remesh & solidify
Alt+G	clear location → object to world origin
Period	pivot pie: 3D Cursor / Individual Origins
0	proportional editing; wheel sizes falloff
Alt+Z	x-ray · C circle select · Ctrl+I invert
L / Ctrl+L	select linked
Ctrl+click 2nd	select everything between two faces

MESH EDITING

Ctrl+R	loop cut (wheel adds cuts)
I	inset — 'Individual' toggle for per-face
J	connect two verts with an edge
Shift+Ctrl+B	VERTEX bevel → chipped-stone corner
X > Dissolve	remove edge, KEEP the face
M	merge: At Center / By Distance
ALT+D	LINKED duplicate — prefer for hardware
P	separate · Ctrl+J join · Shift+N normals

THE ENVIRONMENT MOVES — Ironbark

radial dup	Period → pivot 3D CURSOR → Shift+D, right-click, R Z 90 ×3
Spin tool	front view, Y axis, then Steps 8 / Angle 90 / Centre X,Z
sharp edges	Select > Select Sharp Edges → Mark Sharp → Shade Smooth
Fractal 1.2	Edge > Subdivide with Fractal = instant stylised wear
re-centre	Shift+S Cursor to Selected → Origin to 3D Cursor → Alt+G → Shift+C

ADD-ONS (all ship with Blender — just enable)

LoopTools	right-click > LoopTools > Circle / Loft / Relax
Extra Mesh Obj	Shift+A > Mesh > Extras > Wall Factory
ANT Landscape	Shift+A > Mesh > Landscape — preset CANYON

SCULPT + BLENDER 5

Flatten Contrast	small brush = genuinely SHARP sculpted edges
Box Trim	boolean cut — outperformed ZBrush in live test
Front Faces Only	Tool > Advanced — back-face masking
Vulkan	Preferences > Display Graphics — faster high-poly
Temperature	light colour node, replaces the blackbody hack
remesh voxel	0.01 typical. 0.001 gave 44 MILLION polys.

THE BRIDGE — ZBRUSH ■ BLENDER

The handoff, not a blend. One crossing each way.

ZBRUSH OWNS

mask → extract shells off a body · Dynamesh design exploration · DamStandard / Trim / Slash / Flatten damage · IMM hardware: trim, bolts, stitches, chainmail · layers + morph target · polypaint · ZRemesher with polygroup steering

BLENDER OWNS

the shrinkwrap panel method · vertex-group offsets that stay parametric · exact operations: Spin with known Angle/Centre, Array, Mirror · precise re-centring and origin control · lattice · predictable booleans · non-destructive stacks you can revisit months later

Neither owns 'sculpting' or 'modelling' outright. Route by the OPERATION.

SCALE — the thing that bites

ZBrush is unit-agnostic; Blender is not. Fix ONE convention or the kit won't compose: +Z up, +Y front, humanoid = 1.8 units. Baril's mm values (3 mm plate, 3.5 mm layered sheet, 6 mm deep offset) only mean something once it's fixed.

THE ONE THING THAT DECIDES IF THIS IS PLEASANT

A raw ZBrush Extract is dense, uneven and triangulated. Blender's stack assumes even quads. Feed it a raw extract and every modifier fights you.

SO: ZREMESH BEFORE YOU CROSS. Always.

Extract → Polish By Features → ZRemesher LOW → Polish again. 2–5k quads is a good crossing weight. TriGon already does this inside ZBrush for his own reasons — one step, two payoffs.

THE STRONG MOVE

Use the ZBrush extract as the BASE MESH and shrinkwrap new panels onto it. It already follows the body, it's smooth, and you never edit it again. All the plate detail happens on panels above it.

POLYPAIN SURVIVES

Polypaint is per-vertex colour → lands in Blender as a COLOR ATTRIBUTE. `forge/intake.py` already transfers vertex colours across a remesh by KDTree. Caveat: polypaint needs DENSITY — a 2k crossing mesh can't carry it. Paint after you're back at high subdiv, or bake to texture.

Don't bounce repeatedly. If you're on the third crossing, the piece wants to live in one tool.

LAWS WORTH MEMORISING

...and the keys for your own sculpt tool

THE ASYMMETRY RULE — Abe Leal

The closer a detail is to the CENTRE LINE, the more obvious symmetry becomes — you can never see both far-side details at once. So details near the centre line need deliberate asymmetry; far-side ones can stay mirrored.

THE OVERLAP LAW — Abe Leal

Where two pieces meet, build a deliberate OVERLAP and push the back section further in. Two payoffs: you can retopo the pair as ONE piece, and you get a much better AO bake / cast shadow.

THE THICKNESS LAW — Abe Leal

Always make hard-surface pieces slightly THICKER than reality. Razor-thin edges alias badly in engine and read poorly. 'A little bit chunkier than the real world.'

PANEL SEAMS — bevel then delete

For a TRUE gap between plates: select the real seam edge → BEVEL it very small → DELETE the resulting faces → then Solidify → then Subdivision.

LEATHER IS SKIN — TriGon

When you punch INTO leather, the material beside the punch RAISES. After the indent pass, raise the surrounding area slightly. That's what kills the plastic look.

dc-sculptgl KEYS (v3.2)

From your own forge/README.txt

1-9 / 0	tools
E	transform · X radius
C	intensity · N negative
S	picker · Del delete
F / T / L	views · Space reset
W	wireframe
ADDON	
M	retopo mark
O	iso 35.264° ortho view
A	masking · Q local scale

THE FORGE ENGINES

intake	clean → remesh → vertex-colour transfer
rings	edge rings from painted marks
bake	UV + texture bake
panel	the shrinkwrap armour-panel stack (new)

```
blender -b -P forge/panel_stack.py -- --out panel.blend
```

Verified on Blender 5.0.1: modifier order correct, offset group displaces by exactly the configured amount.